

The CFDE Workbench: Integrating Metadata and Processed Data from Common Fund Programs

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Abstract

The NIH Common Fund Data Ecosystem (CFDE) program was established to facilitate data accessibility and interoperability across multiple Common Fund (CF) programs, promote collaborations and accelerate discoveries by combining diverse data types from different CF programs. The CFDE Data Resource Center (DRC) was tasked with developing two web-based portals: an Information Portal to serve information about the CFDE, and a Data Portal to host harmonized metadata and processed data contributed by participating CF Data Coordination Centers (DCCs) and other sources. To achieve these goals, the CFDE DRC developed the CFDE Workbench, a web-based platform that hosts processed data, metadata, tools, use cases, and analyses developed by the CFDE. The Cross-Cut Metadata Model (C2M2) and several other processed data are hosted by the CFDE Workbench, including set libraries (XMTs), Knowledge Graph (KG) assertions, and attribute tables. These processed data formats make information derived from CF programs more findable, accessible, interoperable, and reusable (FAIR), and artificial intelligence (AI)-ready for cross-DCC knowledge discovery. Besides serving data, metadata, and code assets, the CFDE Workbench has also developed several tools that utilize these resources to enable cross-CF-program knowledge discovery use cases. Overall, the CFDE Workbench is a platform that consolidates efforts toward making CF resources harmonized, FAIR, and AI-ready. The CFDE Workbench website is available from https://cfde.cloud.

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Introduction

The Common Fund Data Ecosystem (CFDE) and the Data Resource Center (DRC)

Since its establishment in 2006, the National Institutes of Health (NIH) Common Fund (CF) Program has supported significant advancements in biomedical research by funding ambitious, cross-institutional initiatives. Examples of CF programs include the Genotype-Tissue Expression (GTEx) [1], the Human Microbiome Project (HMP) [2], Undiagnosed Diseases Network

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(UDN) [3], Knockout Mouse Phenotyping Program (KOMP2) [4], Metabolomics [5], Glycoscience [6], Gabriella Miller Kids First Pediatric Research Kids First [7], Human Biomolecular Atlas Program (HuBMAP) [8], Stimulating Peripheral Activity to Relieve Conditions (SPARC) [9], Illuminating the Druggable Genome (IDG) [10], the Library of Integrated Network-Based Cellular Signatures (LINCS) [11], Bridge to Artificial Intelligence (Bridge2AI) [12–15], Cellular Senescence Network (SenNet) [16], Extracellular RNA Communication (ExRNA) [17], Somatic Cell Genome Editing (SCGE) [18], Cellular Senescence Network (SenNet) [19], Acute to Chronic Pain Signatures (A2CPS) [20], and the Molecular Transducers of Physical Activity Consortium (MoTrPAC) [21]. CF programs have produced diverse high-dimensional datasets ranging from omics profiling of human and mouse cells and tissues to other deep phenotyping data, including imaging and clinical measurements. Many of these datasets are served via online databases and tools that have been widely used by the biomedical research community. Recently, the CF established a new program called the Common Fund Data Ecosystem (CFDE) to further the CF datasets' findability, accessibility, interoperability, and reusability (FAIRness) [22] as well as improve artificial intelligence (AI)-readiness. The goal of the CFDE program is to assist biomedical researchers in forming complex hypotheses that utilize and integrate data generated from several CF programs. The CFDE consortium is made of five centers: the Data Resource Center (DRC), the Knowledge Center (KC), the Integration and Coordination Center (ICC), the Training Center (TC), and the Cloud Working Implementation Center (CWIC). In addition, data coordinating centers (DCCs) from participating CF programs are members of the CFDE consortium. Here we describe the CFDE Workbench, the primary product of the CFDE Data Resource Center (DRC).

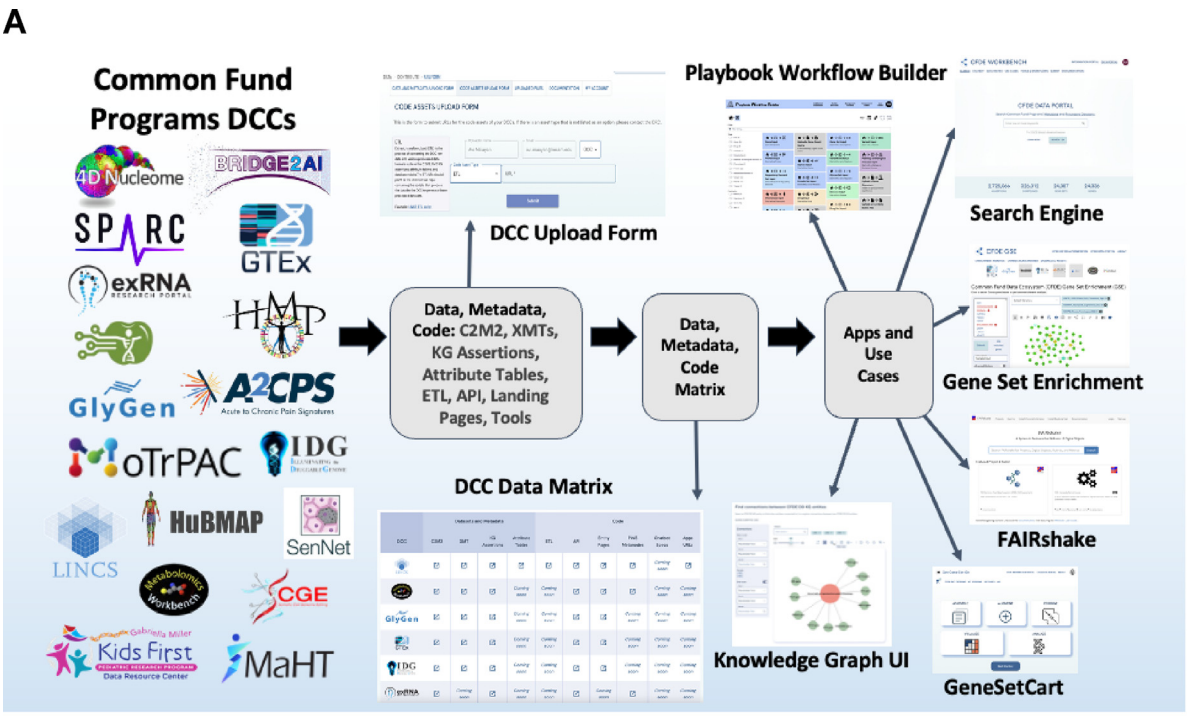
Methods and Results

The CFDE Workbench workflow

The CFDE Workbench is organized into two main components, the Data Portal (https://data.cfde.cloud/) and the Information Portal (https://info.cfde.cloud/). The Data Portal catalogs several types of uniformly processed data, metadata files and other digital assets from each participating DCC. These files and assets are submitted to the portal by the DCCs and can then be downloaded from the Data Matrix or searched for via the Data Portal search engine. The Information Portal provides relevant information about each DCC and on CFDE consortium activities, including training and outreach activities, publications, brief descriptions of CFDE partnership projects, and detailed community-established protocols. Here, we describe the various components of the CFDE Workbench and how they all come together to produce a valuable and comprehensive resource for the CFDE consortium and the biomedical research community at large (Figure 1).

The CFDE Workbench submission system

The CFDE Workbench functions as a hub for finding CF-supported datasets via a dedicated search engine. Digital assets on the CFDE Workbench are broadly grouped into two categories: file assets and code assets. File assets include Cross-Cut Metadata Model (C2M2) [23] packages, knowledge graph (KG) assertions, set libraries (XMT), and attribute tables. Code assets include API documentation links, entity page templates, URLs to apps, extract, transform and load (ETL) scripts, AI Assistant specifications, URLs to tools, and models (Table 1). To streamline the collaborative efforts of the CFDE and to ingest these assets into the CFDE Workbench, the DRC has implemented a data and metadata submission system. This system enables the DCCs to submit their assets to the CFDE Workbench. The submission system assigns users to one of four roles, each having different access permissions to various pages within the system: a general user, a DCC submitter, a DCC approver, and a DRC reviewer. The system has two user-friendly input forms: 1) the upload form, designed for uploading data and metadata files; and 2) the code assets upload form for submitting URLs and descriptions of the various code assets. By using the data and metadata submission system, DCCs can upload and approve their assets for public display on the CFDE Workbench. Upon submission of a new asset, DCC submitters and approvers can use the system to archive their previously published assets and mark their most up-to-date assets as current. In addition, once assets are uploaded, these assets are evaluated for FAIRness with FAIRshake [24] (Figure 2). Different FAIRshake rubrics have been established to evaluate each asset type. The results from the automated FAIRshake FAIR assessments are displayed as an insignia near each uploaded asset entry. These FAIRshake insignias are only visible to the data submitters and approvers from each DCC, and they can only see the insignias pretaining to their uploaded assets. The FAIR assessments aimed to provide the DCC submitters and approvers with feedback on how they may improve the FAIRness of their uploaded assets. Comprehensive documentation is provided to assist submitters and approvers in preparing, submitting and approving them. The documentation contains details about the asset types, the submission and approval process, and descriptions about how to prepare the asset types for submission. Overall, by contributing these assets to the workbench, the DCCs significantly enhance the CFDE and make the assets FAIR and more valuable.



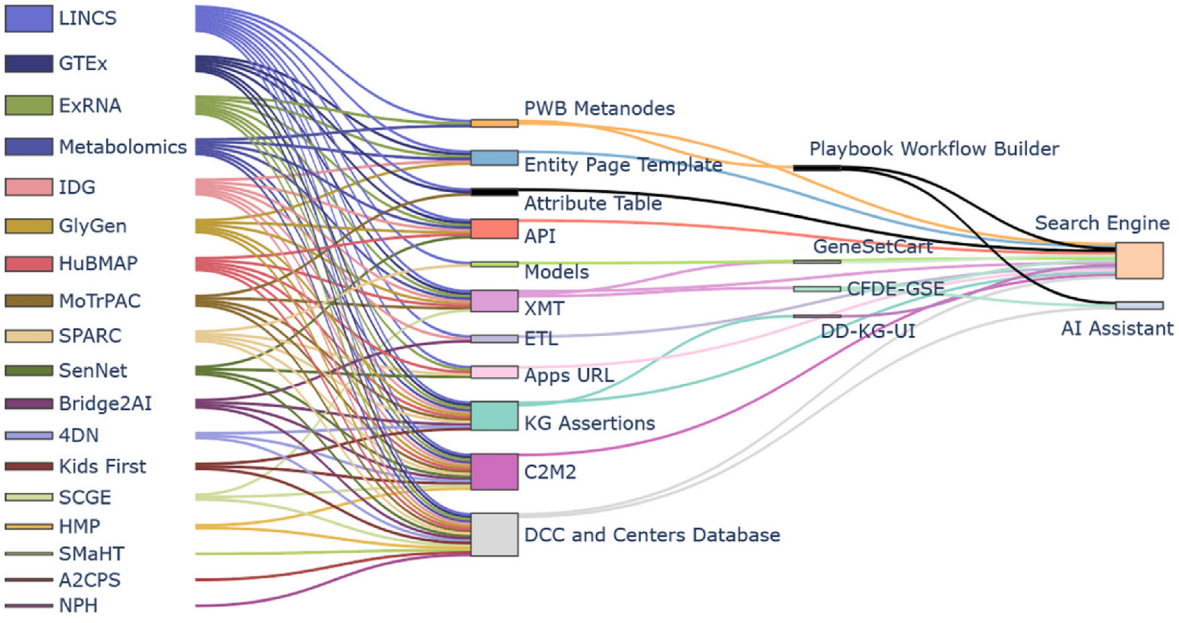


Figure 1. (A) The CFDE Workbench receives metadata, processed data, and code assets from the CF program participating DCCs. These assets are uploaded via a data submission system and assessed for FAIRness with FAIRshake. Approved uploads are displayed in a Data Matrix and served via a search engine. The metadata, processed data, and code assets enable the development of a layer of tools that include the PWB [44], GeneSetCart [47], DD-KG-UI [46], and CFDE-GSE. (B) Sankey diagram demonstrating how processed data and metadata items from the various DCCs flows into the CFDE Data Portal and then feed the CFDE Workbench tools, the search engine and the AI Assistant.

Table 1 Summary of metadata, processed data, and code assets that are served by the CFDE Workbench and displayed within the Data Matrix.

Asset type	Data Type	Description	Format
Datasets and Metadata Assets	Cross Cut Metadata Model (C2M2)	A metadata standard for describing data files from a variety of biomedical and biological sources.	zip (tsv)
	XMT	Text-based files that contain a collection of sets of an entity type, for example, gene set libraries stored in a gene matrix transpose (GMT) format.	gmt or dmt
	KG Assertions	Nodes and edges that describe relationships between different biomedical concepts, objects, and entities.	Zip (csv)
	Attribute Tables	Files containing tables that describe the relationship between two entities with one entity type on the rows (for example, genes) and another on the columns (for example, tissue types). The table is filled with values that quantify the association between the entities.	h5 or hdf5
Code Assets	ETL	Extract, transform, load (ETL) scripts that process raw data into various processed data formats such as the C2M2, XMT, KG assertions, attribute tables, and database tables. URL redirects to the DCC GitHub repository, which contains the scripts that DCC uses to generate these processed datasets.	URL, py
	API	REST API to access DCC data and tools. URL redirects to the API documentation. The API recommended standard is OpenAPI, and SmartAPI is the recommended repository for depositing the API docs.	URL
	Entity Pages Template	A template used to create the landing page displaying the datasheet about a gene, a metabolite, a protein, a cell type, or other biological and biomedical entity.	URL
	PWB Metanodes	The components of the PWB [44] that can receive inputs and produce outputs, including visualization and data processing operations. URL redirects to a GitHub link that contains the script describing a PWB metanode.	URL (tsx, json)
	Models	Models are loosely defined and can fit many different types of computational constructs that do not fit as tools or workflows. These models can be AI/ML models, dynamical models of cells, molecules, or tissues, such as physiological ODE models.	URL

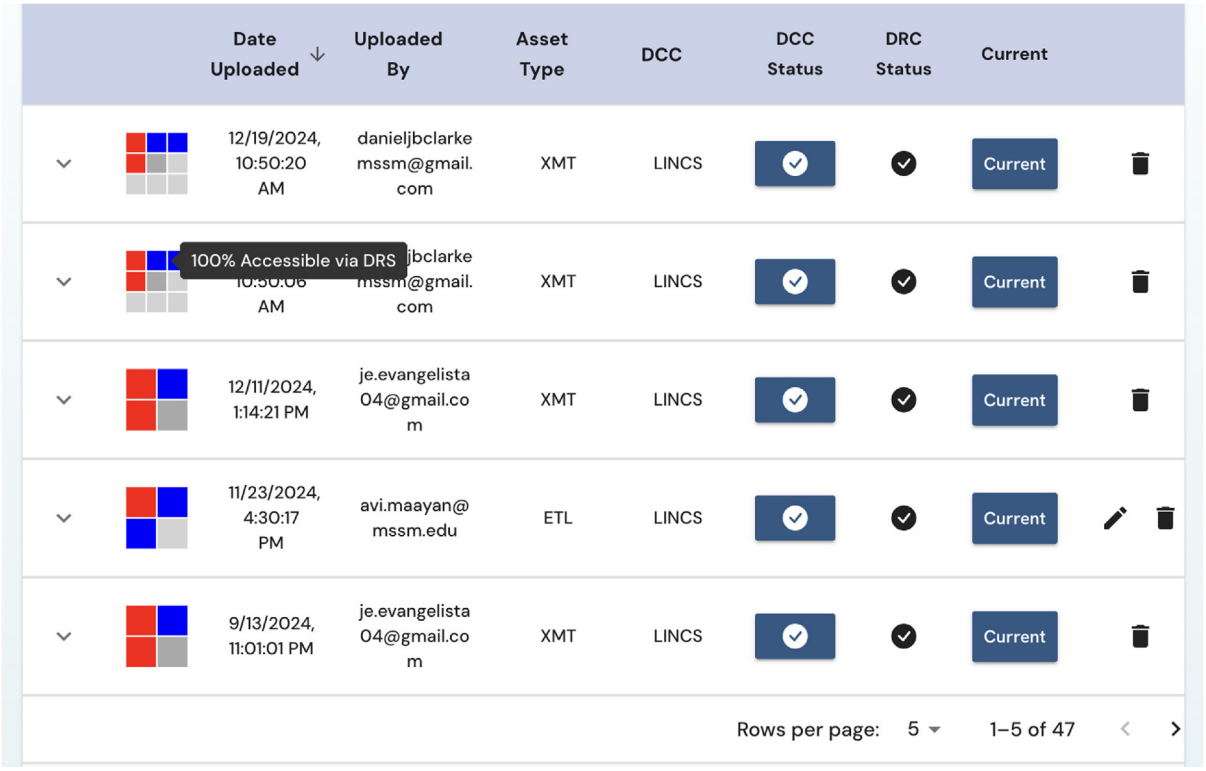


Figure 2. Screenshot from a page within the data submission system that displays a list of assets uploaded by a DCC. Near each uploaded listed asset, the FAIRshake [24] insignia displays the results of the FAIR evaluation of the asset. The table also has interactive switches to approve uploaded assets by the DCCs and the DRC.

Metadata packages for capturing common metadata elements

The Cross-Cut Metadata Model (C2M2) was established to ensure data discoverability across the CF resources [23] and has evolved with the addition of newer CF programs into the CFDE. C2M2 packages consist of metadata items organized in approximately 50 related tables. For example, these tables contain records about DCCs, projects, collections, subjects, biosamples, files, anatomy, disease, and other descriptive metadata elements (Figure 3). The use of ontologies and controlled vocabularies is a key feature of the C2M2, ensuring the harmonization of terms within and across DCCs and their assets. Some of these ontologies and controlled vocabularies include Uberon [25] for anatomy, InterLex [26] for biofluid, Human Phenotype Ontology (HPO) for phenotype [27], UniProtKB [28] for protein, Ensembl [29] for gene, bioscientific data analysis ontology (EDAM) [30] for file format and data type, Ontology for Biomedical Investigations (OBI) [31] for assay type and sample prep method, PubChem [32] for compound and substance, and Disease Ontology (DO) [33] for disease. The first step in the C2M2 ingestion pipeline is to ingest the metadata from the tab-separated value (TSV) files into a database.

To facilitate meaningful and fast full-text queries across all these tables, the CFDE Workbench utilizes the foreign-key constraints among these tables. These keys are joined to produce a fully flattened (FFL) table that is indexed with respect to multiple columns, such as anatomy, species, disease, assay type, and data type. The FFL table is centered around biosamples which are the most fine-grained entity in the collection of metadata items across the CFDE DCCs. As of December 2025, the CFDE Workbench contains approximately 2 million biosamples. Each biosample comes with additional fields such as anatomy. A summary of top datatype, file types, files, and biosamples are provided (Tables 2 and 3). To incorporate key metadata from the files table ($\sim$6 million rows) in the search and filtering operations, distinct project, assay type and data type fields are saved in another table. This additional table is incorporated into the FFL table. Further, a searchable column of the tsvector data type that combines information from all the key columns, is added to the FFL table to facilitate the query supporting the search. In addition to metadata packages, processed data packages such as XMTs and KG assertions can be ingested into respective tables in the database to facilitate complex queries.

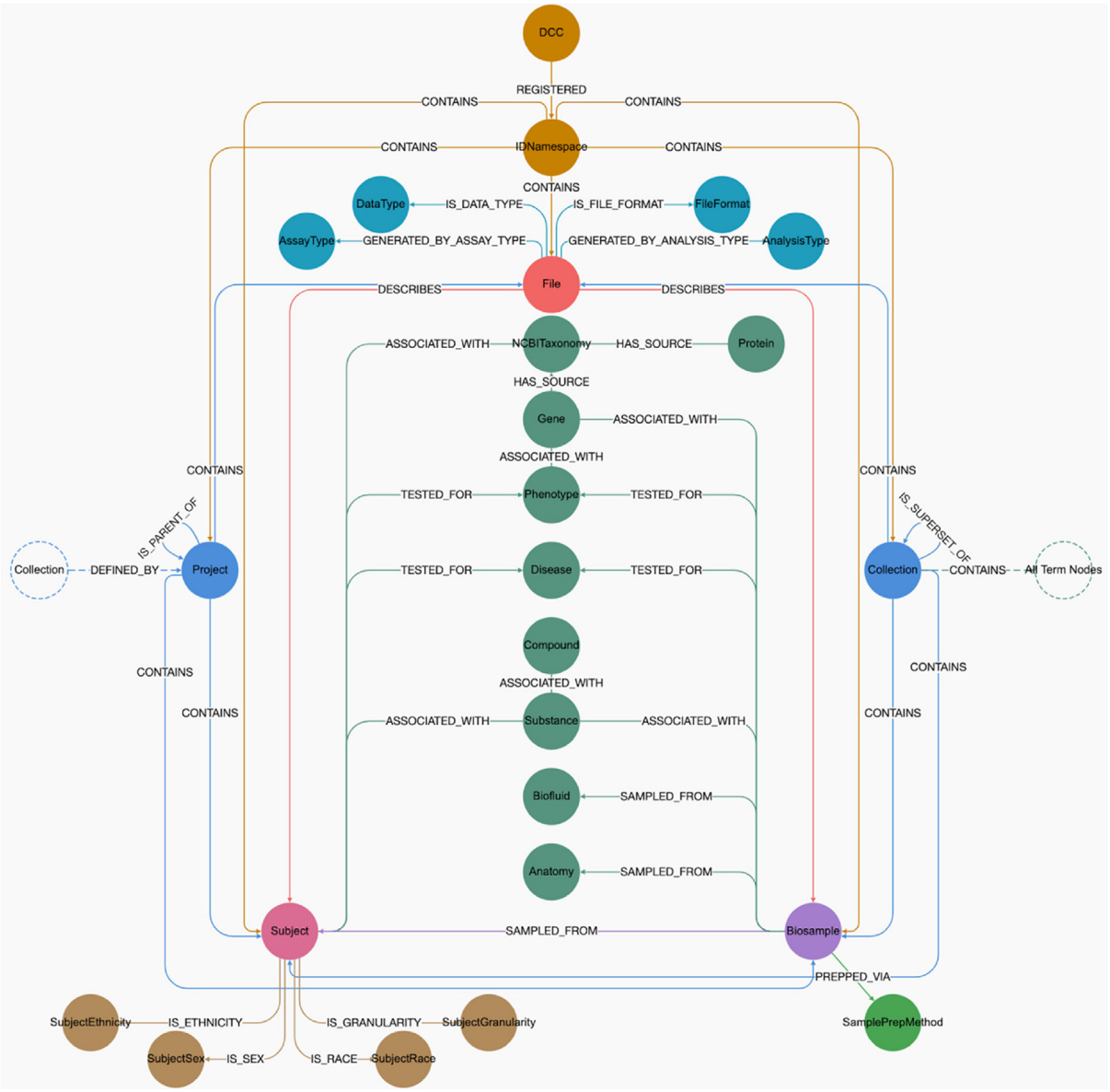


Figure 3. The C2M2 schema diagram illustrates foreign key relationships between tables.

Table 2 Top C2M2 data field values across the consortium as of Dec 2025.

Table in C2M2	Field	Unique	Total
file	analysis_type	5	49,100
biosample	sample_prep_method	8	1,472,359
biosample	anatomy	391	1,680,056
file	data_type	40	3,023,964
file	mime_type	69	4,632,739
file	file_format	56	5,458,215
file	assay_type	98	6,012,834

The CFDE Workbench Data Matrix

The CFDE Workbench Data Matrix (https://data.cfde.cloud/matrix) is a key component of the CFDE Workbench Data Portal (Figure 4). The Data Matrix catalogs available assets from across CF programs. The following 18 CF participating programs contributed, or plan to contribute, to the Data Matrix: MoTrPAC [21], GlyGen [6], Kids First [7], A2CPS [20], Metabolomics [5], LINCS [11], SenNet [19], GTEx [1,34], Human Microbiome Project (HMP) [35,36], 4D Nucleome (4DN) [37,38], ExRNA [17], SCGE [18], SPARC [9], HuBMAP [8], IDG [10], Bridge2AI [13,15,39,40], Somatic Mosaicism across Human Tissues (SMaHT) [41], and Nutrition for Precision Health (NPH). Although individual CF DCCs remain responsible for hosting the original data produced by their respective programs, the CFDE Workbench Data Matrix enables users to access standardized and abstracted repre-

Table 3 Top C2M2 table records across the consortium as of Dec 2025.

DCC	File	Biosamples
HMP	251,136	51,873
LINCS	1,495,871	1,466,796
MoTrPAC	18,980	6156
4DN	53,697	4406
GTEx	665	22,015
SCGE	0	1885
HuBMAP	2,990,687	3616
SenNet	0	2480
Kids First	1,101,617	110,904
Metabolomics	6092	380,326
IDG	758,93	0
GlyGen	2021	0
SPARC	125,536	6160
Total	**6,122,195**	**2,056,617**

sentations of knowledge derived from the raw data of different CF programs in one location. The Data Matrix also tracks provenance, performs FAIR assessments on the submitted files, and facilitates DCCs to control asset visibility. The Data Matrix additionally supports integration with external community standards such as the OpenAPI [42] and SmartAPI [43] initiatives, and with internal CFDE efforts to develop original standards such as Playbook Workflow Builder (PWB) metanodes [44], Model Context Protocol (MCP) server specifications [45], and continually extending the C2M2 to support other metadata types. By making CF DCC knowledge available for download in community established standardized formats, the CFDE Workbench Data Matrix aims to catalyze usage and increase the relevance of CF data and metadata assets within the CFDE and beyond.



Figure 4. Screenshot from the CFDE Workbench Data Matrix.

The CFDE Workbench Data Portal search engine

The CFDE Workbench Data Portal search engine was developed to enable user-friendly queries of CF programs' highly processed data, metadata, as well as tools and analysis platforms developed by the CFDE, and by the participating DCCs (https://data.cfde.cloud/). To expand the FAIRness of CFDE resources, the CFDE Workbench search engine serves the ingested C2M2 packages [23] and processed data formats, including XMTs, Knowledge Graph (KG) assertions, and attribute tables, through a unified search interface. These processed data formats make information derived from the underlying CF programs readily accessible with the aim of easier cross-DCC data integration. Besides serving data and metadata, the search results are linked to downstream bioinformatics tools for data analysis. The CFDE Workbench search engine supports search syntax. For example, entering two search terms together (term1 term2) works as an AND operator, while adding the word OR between search terms (term1 OR term2) returns the union of searching each term individually. The dash symbol (-term1) can be used to exclude a specific search term, and the double quotes ("term1 term2") preserve the ordering and consider the search as a bigram. The processed data and metadata formats derived from the DCCs are assembled and incorporated into the unified database through a series of scripts. Assets are registered to dedicated tables and linked to a common table indexed for full-text search when applicable. The C2M2 metadata standard describes experimental data across the CF DCCs, linking biosamples and files together with standard ontologies [23]. The C2M2 metadata is ingested into the database, and several metadata views are made searchable, for example, raw data files cataloged by the DCCs.

The XMT files ingested into the Data Matrix are sparse matrix serializations of set libraries, where the X refers to the elements in the sets. For example, a Gene Matrix Transpose (GMT) is a library of gene sets, and a Drug Matrix Transpose (DMT) is a library of drug sets. Searchable terms that describe sets are stored in the database and linked to the set of entities defined in the XMT record. KG assertions are triples that define a typed relationship between two typed nodes from the Data Distillery (DD) KG [46]. Searchable typed nodes from the KG assertions and typed relationships are stored in the CFDE Workbench database, and these assertions are linked to a record dedicated to the assertion. Where applicable, records are associated with the source data file and the source DCC. The database is queried by a Next.js web server that provides an interface for listing DCC assets. Searching for assets is based on their description and other content found within the processed data assets. Search results can be filtered by asset type or by DCC. A search query returns C2M2 metadata (Figure 5A) and processed data search results (Figure 5B). A C2M2 metadata search returns projects based on the query term. Users can refine the results by disease, species, anatomy, gene, protein, compound, data type, assay type, biofluid, post translational modification (PTM), and DCC. The whole or selected query result can be downloaded in JSON format containing project details and metadata information. The respective project page, also referred to as the record page, has additional context such as species, anatomy, and disease. The page also provides information about the related biosamples, subjects, collections, and files, some with persistent IDs and access URLs. All or part of this information can be downloaded in JSON format. The processed data search returns results based on the entity type. The entity types include genes, gene sets, diseases, anatomies, proteins, phenotypes, and more. These entities have dedicated pages that list information about each entity, including links to tools for further analysis, such as finding more information about a gene or performing enrichment analysis on a gene set. The CFDE Workbench provides CFDE-developed tools to help users further investigate knowledge about the entity of interest. For example, when a user submits a query for a specific human gene name (Figure 5C), several CFDE tools for analyzing the gene are returned in the search results page as clickable cards. Selecting a tool redirects the user to the tool with the gene loaded and analysis ready to be viewed.

The CFDE Workbench AI assistant

The advent of large language models (LLMs) presents the opportunity to enhance the experience of CFDE Workbench users by implementing AI assistants. We have implemented such an AI assistant for the CFDE Workbench (https://data.cfde.cloud/chat) (Figure 6). This assistant sends user queries to GPT-4 using the OpenAI Responses module (https://platform.openai.com/docs/api-reference/responses) which retrieves information from a connected Model Context Protocol (MCP) [45] server that acts as an intermediary between the model and CFDE resources (Figure 7). These resources include information about the CFDE Centers and participating Common Fund Programs. User queries can range from questions about publications and outreach activities to how to integrate data from different CF programs. Additionally, the MCP server provides the assistant with access to standardized workflows that draw knowledge from multiple CF programs and other resources using the Playbook Workflow Builder (PWB) [44]. The PWB provides a simple mechanism for exporting workflows into a programmatic query which can then be executed through the PWB API. The MCP server enables the chatbot to retrieve these workflows for

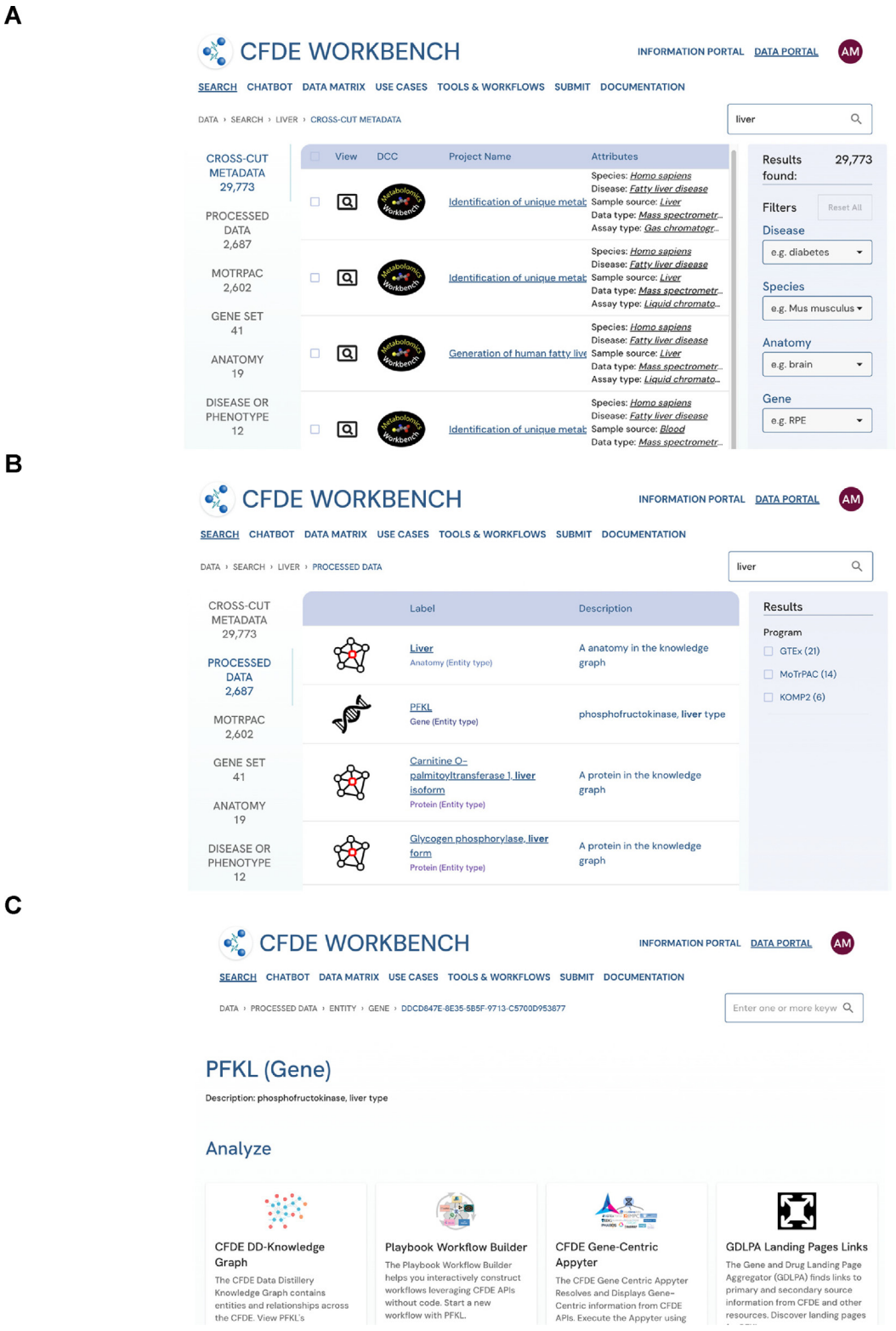


Figure 5. Screenshot from the CFDE Workbench search engine. (A) Results from the C2M2 search for the search term “liver”. (B) Results from the processed datasets search for the search term “liver”. (C) Gene landing page for the gene PFKL.

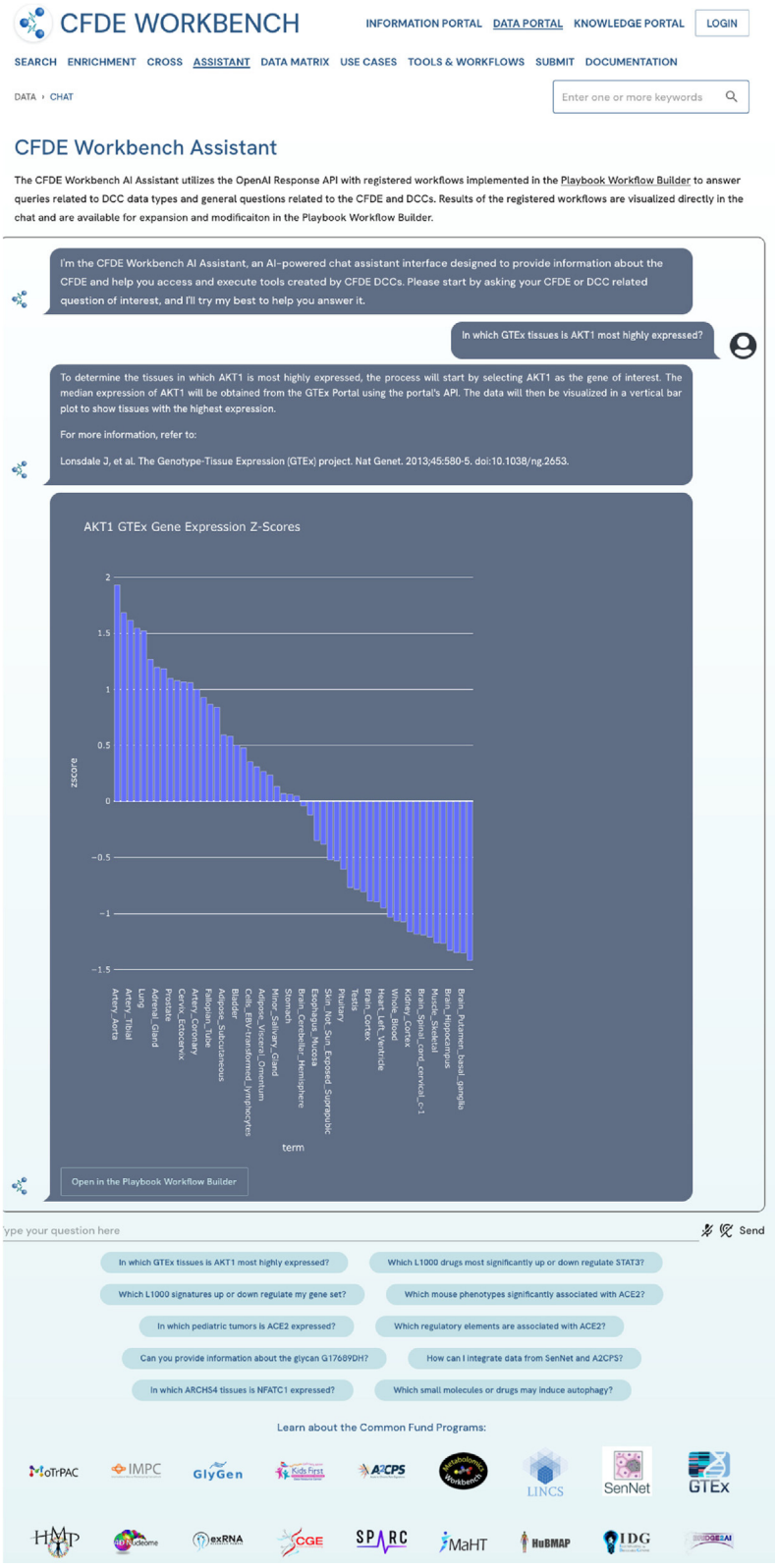


Figure 6. Screenshot from the CFDE Workbench AI Assistant interface. The results shown are from asking the chatbot to provide the expression levels of the gene ACE2 across tissues using the GTEx API.

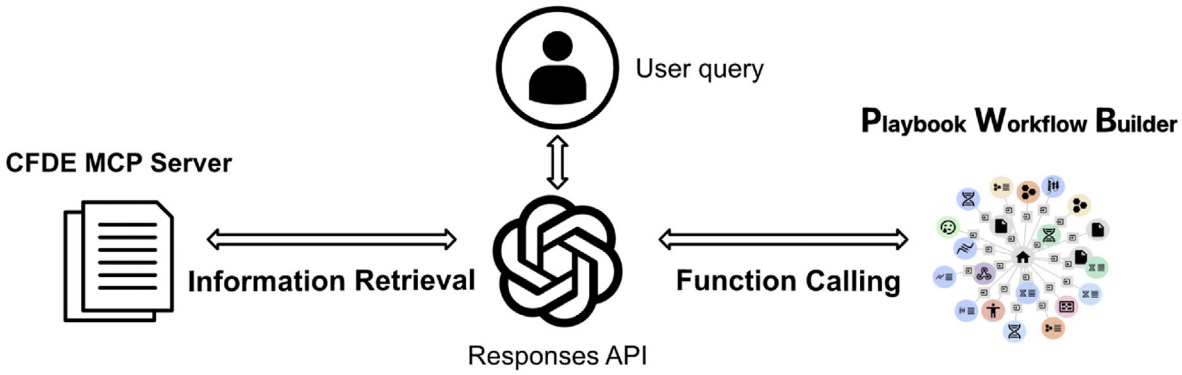


Figure 7. The CFDE Workbench AI Assistant uses OpenAI's Responses API to send user queries to GPT-4. The LLM then queries the CFDE MCP server and selects an appropriate tool based on the query. These tools provide information about the CFDE, the different centers, and the different Common Fund Programs involved in the consortium. Additionally, there are tools that define several workflows from the Playbook Workflow Builder that utilizes and integrates data from the different CF Programs and are available for users to interact with via the chat interface.

users to run processes directly in the assistant chat interface. Once a workflow is retrieved, frontend components adapted from the PWB metanodes are then used to visualize the results returned from the workflows invoked via the assistant. In addition to providing such visualizations directly in the AI Assistant interface, the AI Assistant provides a link to the executed PWB workflows in the PWB interface for modifications and expansion. New tools can be added to the server as the consortium grows and new workflows are developed. To evaluate the performance of several LLMs for their ability to invoke the right function calls, and provide the most complete and accurate answers, we benchmarked several LLMs. The LLMs that were tested are: GPT-3.5, GPT-4, GPT-5, gpt-oss:20b, gpt-oss:120b, Mistral Large 3, groq-llama-3.3-70b-versatile. We provided each LLM the collection of example questions provided below the AI Assistant chat interface, and then manually evaluated the replies. The evaluation was conducted by a cohort of human reviewers who blindly scored each response. All example questions that are displayed below the AI Assistant on the CFDE Workbench site were submitted to all the above listed LLMs and reviewed by eight human reviewers. The results from such a survey suggest that GPT-5 outperfomed Mistral, GPT-4, and the rest of the tested LLMs (Figure 8). Overall, the CFDE Workbench AI Assistant provides users with an alternative useful way to obtain information about CF programs, and to perform analyses using CF datasets and API. To mitigate "hallucinations", a system prompt provides directions to strictly use information from the MCP server when answering queries. Additionally, the assistant is instructed to only answer questions related to the CFDE, the CF, and the DCCs. Additionally, by using the previous_response_id parameter, an individual user can have threaded conversations with the model. This feature enables follow-up questions that consider prior context for improved user experience.

CFDE Workbench downstream analysis tools

Organizing the data, metadata, and APIs from a collection of CF programs facilitates downstream analysis tools that draw from these resources to enable hypothesis generation that combines CF datasets and tools from multiple programs. So far, these downstream analysis tools and platforms include the PWB [44] (https://playbook-workflow-builder.cloud/), GeneSetCart [47] (https://geneset-cart.cfde.cloud/), CFDE Gene Set Enrichment (CFDE-GSE) (https://gse.cfde.cloud/), Gene and Drug Landing Page Aggregator (GDLPA) [48], FAIRshake [24], and CFDE-DD-KG-UI (https://dd-kg-ui.cfde.cloud/) [46]. To demonstrate the utility of the PWB, GeneSetCart, CFDE-GSE, and CFDE-KG-UI, the CFDE Workbench has a dedicated section that provides linkable cards to 25 use cases created with these downstream tools and platforms (https://data.cfde.cloud/usecases). In addition, functionality from CFDE-GSE and GeneSetCart is embedded within the Data Portal. Specifically, users can submit gene set queries against gene sets created from eight CF programs for enrichment analysis with CFDE-GSE under the ENRICHMENT tab (Figure 9A) and cross the same gene sets using the CROSS feature of the workbench (Figure 9B). GDLPA [48] has links to 53 gene, 18 variant, and 19 drug databases that host landing webpages about single gene, variant, and drugs including those produced by the CF DCCs. Users can search by gene, variant, or drug terms and the sites that contain knowledge about these entities are displayed as clickable cards. The FAIRshake platform [24] can be used to assess the level of FAIRness of digital objects by matching each object with a set of questions (FAIR metrics). The

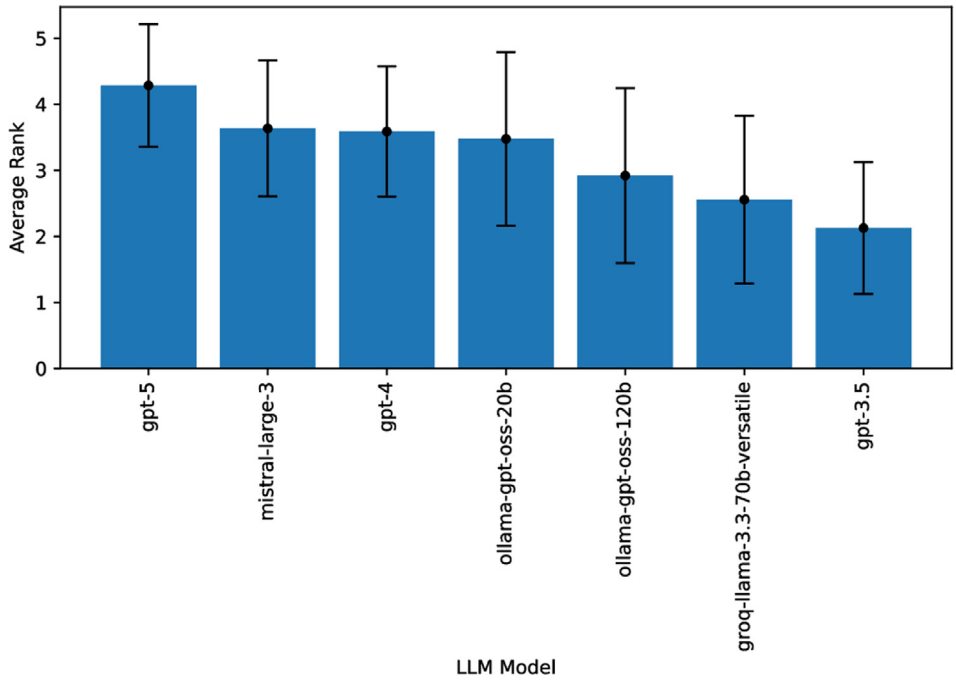


Figure 8. All example questions were submitted to each LLM and then organized randomly into an online survey where responses were scored for answer between 0–5 (5-best). Eight human reviewers evaluated the answers independently.

collection of answers to these questions is visualized as an interactive insignia. FAIRshake is used to provide the CFDE participating DCCs with feedback about the level of FAIRness of their submitted assets. The CFDE data distillery knowledge graph user interface (CFDE-DD-KG-UI) [46] is a web interface that interacts with data collected for a CFDE Data Distillery partnership project. The CFDE-DD-KG-UI leverages the Cytoscape network visualization library Cytoscape.js [49] to visualize queries against the knowledge graph that is stored in a Neo4j database. The DRC developed a generic UI template that can be used to rapidly deploy bioinformatics applications stored in KG databases [50]. GeneSetCart is a platform that helps users of the CFDE Workbench manage and analyze collections of gene sets [47], and CFDE-GSE is described in more detail below.

The CFDE Workbench Gene Set Enrichment (CFDE-GSE) tool

To further enable discovery across CF programs, the DRC systematically processed CF datasets into gene set libraries. These gene set libraries are served from the Data Matrix and from CFDE-GSE (https://gse.cfde.cloud/). CFDE-GSE stores gene set libraries created from the CF programs' datasets as KG in a Neo4j database. CFDE-GSE then hosts these libraries via a web-based application that serves these gene sets for integrated enrichment analysis. So far, 10 gene set libraries have been created by processing datasets from 8 CF programs: LINCS [11], GTEx [1], Metabolomics [5], IDG [10], GlyGen [6], KOMP2 [4], MoTrPAC [21], and HuBMAP [8]. It should be noted that KOMP2, although a CF program, was never an active member of the CFDE. To use CFDE-GSE, gene sets obtained from omics experiments can be submitted to the tool using a simple input form (Figure 10A). CFDE-GSE uses the Fisher's exact test to assess the level of intersection between the input gene set and all gene sets wrangled from the 8 CF programs. The resulting subgraphs produced by CFDE-GSE illuminate connections between the input gene set and various terms describing CF gene sets that significantly overlap with the queried gene set. Such enrichment analyses enable users to explore CF datasets to further hypothesis generation. Besides enrichment analysis, CFDE-GSE can be used to query the consolidated CF knowledge by searching for cross-CF-dataset associations for a gene or a biological term. Users can query the tool to find cross-DCC information about a drug, a metabolite, a tissue, a cell type, or a phenotype. This feature of CFDE-GSE can be used to discover connections between entities that may not be obvious. Future releases of CFDE-GSE would extend the collection of gene sets to serve data from more CF programs as well as add predicted associations using graph-based machine learning methods, opening the door to broader cross-CF knowledge discovery.

Use Case 1: Finding drugs to modify disease genes that change in expression with age

By combining data from the CF programs LINCS [11] and GTEx [1], together with genetics and genomics information about disease genes from sources such as GWAS Catalog [51], ClinVar [52], OMIM [53], and DisGenNET [54], users of the CFDE Workbench can prioritize potential biomarkers and novel therapeutics for many complex human diseases. For example, genes with mutations that are linked to an increase in propensity to develop neurological disorders can be queried against GTEx to assess how their expression levels are changing with age across human tissues. If these genes increase or decrease significantly with age, and play a key role in the disease, the change in expression could be a trigger to promote a disease state. Then, we can use the LINCS resource to find drugs and preclinical compounds that would maximally reduce or increase the expression of these genes to correct their expression to a desired level.

For example, Alzheimer's Disease (AD) is a common brain disorder that typically manifests in the elderly and becomes progressively worse over a span of many years. Symptoms include severe memory loss and dementia. The disease is characterized by abnormal buildup of beta-amyloid plaques and tau tangles, which are protein aggregates found in post-mortem patients' brains [55]. Some forms of AD are familial and have an earlier onset with a specific genetic underpinning. However, for most late-onset cases, there are no clear genetic causes or effective diagnostic biomarkers. To generate hypotheses about potential therapeutics for AD using the tools and datasets

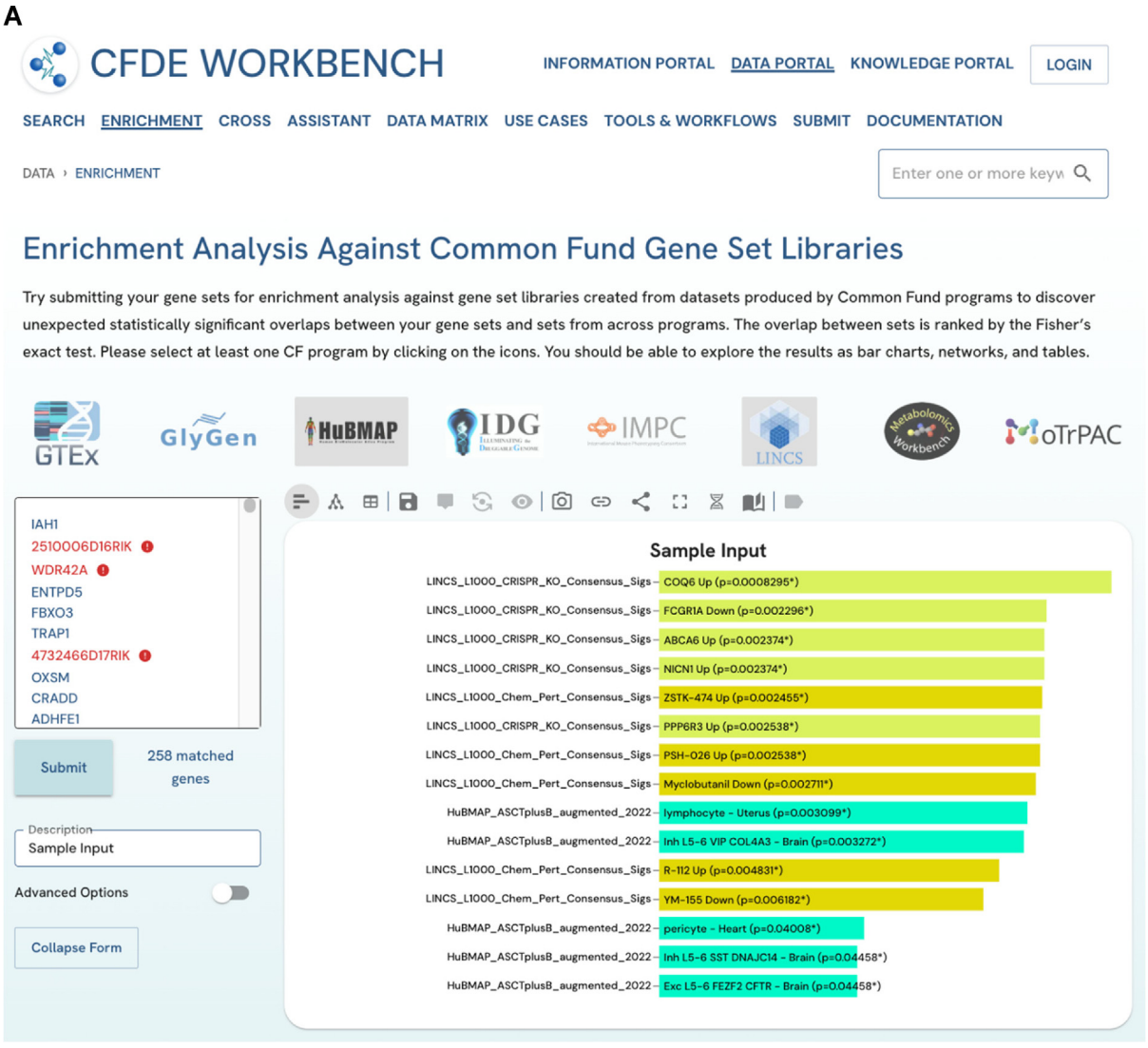


Figure 9. Screenshots from the CFDE ENRICHMENT and CROSS tabs. (A) Results when using the default example gene set and selecting the HuBMAP and LINCS gene set libraries. (B) Crossing results when selecting the GTEx aging signatures library and the MoTrPAC library.

B

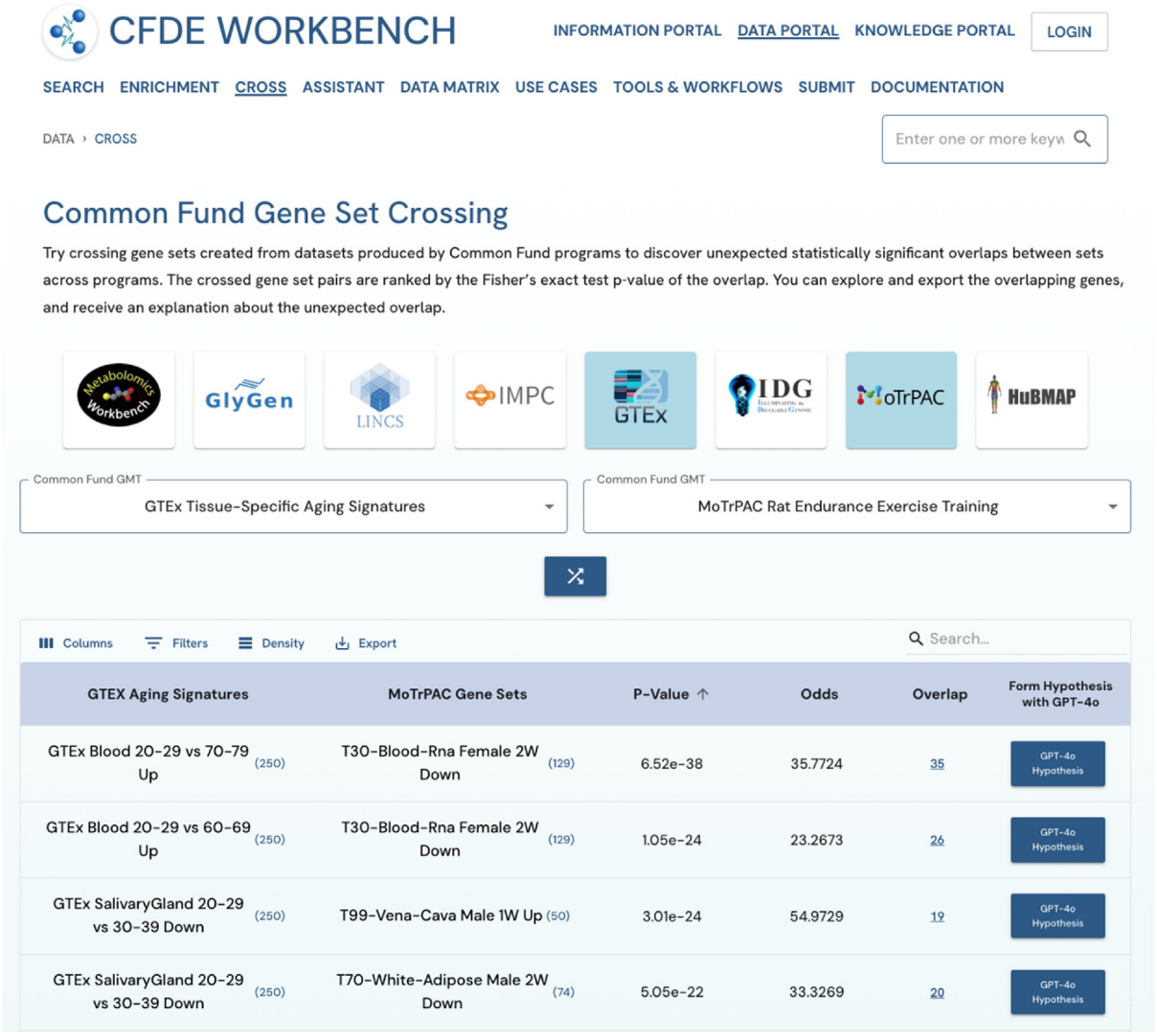


Fig 9. *(continued)*

available from the CFDE Workbench, we begin our analysis with the alzGene database [56], which lists 130 variants in 70 AD causal genes. We submitted these 70 genes for enrichment analysis against the GTEx aging gene set library in CFDE-GSE. The enrichment analysis results show 6 genes that are elevated in blood in the 60–69 age group compared with the 20–29 age group (Fisher's exact test, q-value < 0.01) (Figure 10A). These genes are PLAU, PON1, APOC1, OLR1, APOB, and MPO. Next, using the CFDE AI Assistant, we asked the assistant to identify LINCS drugs that would maximally decrease the expression of these genes one-by-one (Figure 10B). This analysis can also be performed using the PWB [44]. When asking the CFDE AI Assistant, "which LINCS drugs maximally decrease the expression of the gene PLAU?" the top listed drug is NXY-059. Behind the scene, this response is coming from querying the LINCS L1000 dataset [57] with SigCom LINCS [58]. NXY-059 is a drug candidate for treating acute ischemic stroke that works by trapping free radicals to protect the brain [59]. While it showed promise as a neuroprotectant in animal models by reducing brain damage, extensive clinical trials in humans did not show any benefit compared to a placebo and the drug was found to be ineffective for acute ischemic stroke. However, this compound may exhibit long-term benefits for other brain disorders such as AD. Thus, the CFDE Workbench suite of tools enables biomedical researchers to access various resources through a single unified interface to rapidly explore hypotheses about potential therapeutics with genetic and aging underpinnings.

These tools not only bring these datasets together, but also provide means to discover unexpected associations that can lead to novel discoveries.

Use Case 2: Discovering unique information about a gene from common fund datasets

By wrangling, harmonizing, and processing data about genes and proteins from across many CF programs, the CFDE Workbench provides digested and cleaned insights about all human genes and proteins. Such information might be available at each of the CF data portals, but it is impractical for users to visit each of these resources and manually aggregate all the rich available information. The CFDE Workbench brings all this information to the surface providing a unique resource about what is known about a specific gene or protein based on data collected by CF programs.

For example, the effects of mutations in the gene isocitrate dehydrogenase ($NADP^+$) (IDH1) span epigenetic, differentiation, and metabolic programs. Such mutations are highly prevalent across a variety of cancer types, such as low-grade glioma [60], secondary glioblastoma [61], acute myeloid leukemia [62], cholangiocarcinoma [63], chondrosarcoma [64], sinonasal undifferentiated carcinoma [65], and angioimmunoblastic T cell lymphoma [66]. IDH1 mutations are known to occur early during tumorigenesis and are found to have uniform expression in cancer cells, thus making IDH1 a potentially effective therapeutic target. A biomedical researcher interested in investigating the role of IDH1 in cancer and other biological contexts can be aided by the CFDE Workbench, including the search engine, the AI Assistant, and various tools. Rich information about IDH1 can be found by typing "IDH1" in the Data Portal search engine (Figure 11A). The results identify 149 C2M2 specified

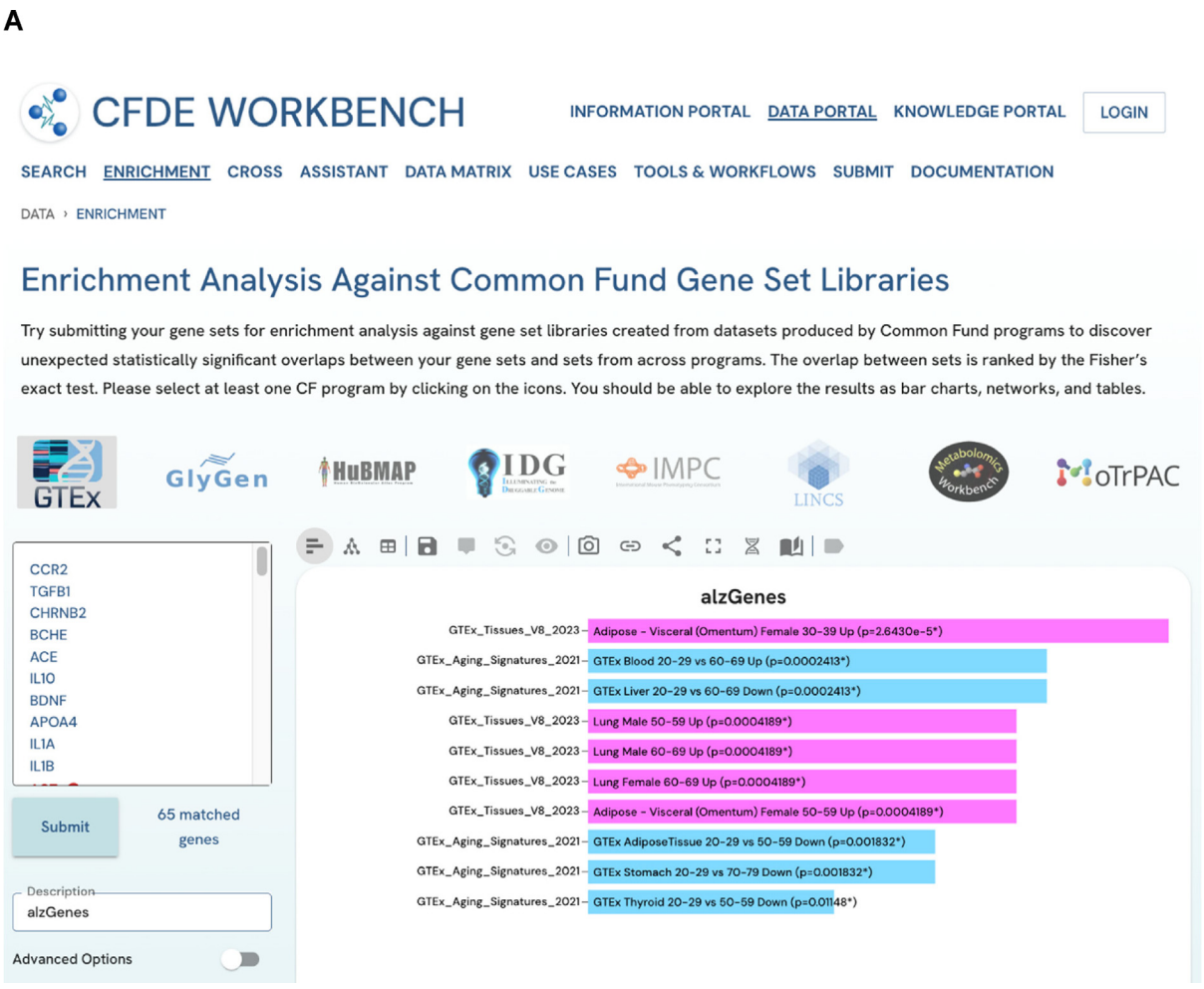


Figure 10. (A) Screenshot from the ENRICHMENT feature of the Data Portal when submitting the 70 Alzheimer's associated genes from the alzGene database and selecting the GTEx libraries. (B) The CFDE AI Assistant is queried for suggesting drugs from the LINCS L1000 data to maximally decrease the expression levels of PLAU which is an Alzheimer's gene that increases in expression with age.

B

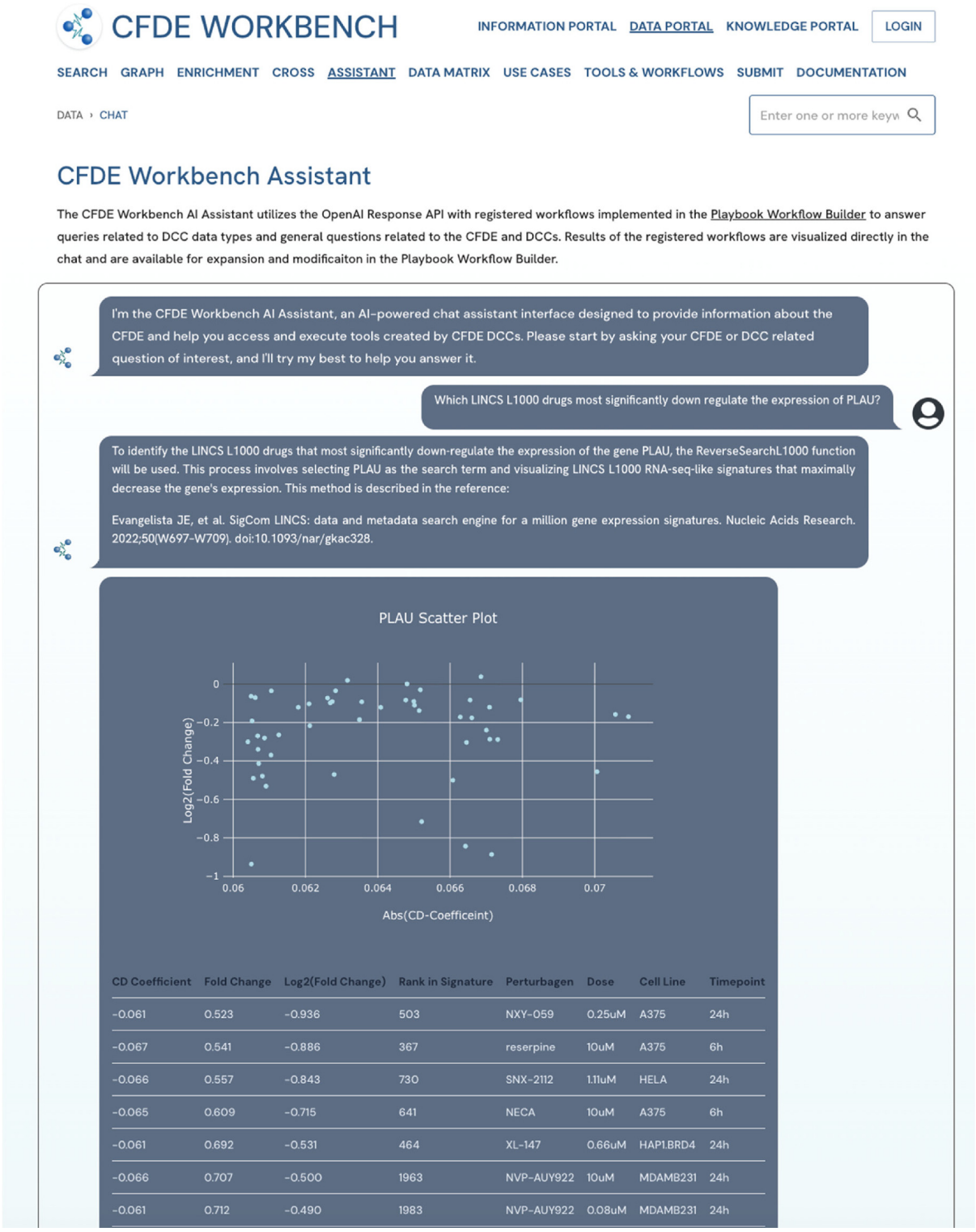


Fig 10. *(continued)*

files from four CF programs: MW [5], LINCS [11], IDG [10] and GlyGen [6]. For example, the MW entries are mostly links to metabolite profiling of cancerous tissues with mass-spectrometry studies. The search also identifies 84 gene sets where IDH1 is included in their description. These gene sets are mostly up- and down-regulated differentially expressed genes computed from the LINCS L1000 data [57]. The search also returns a link to the CFDE Workbench IDH1 gene page. This page

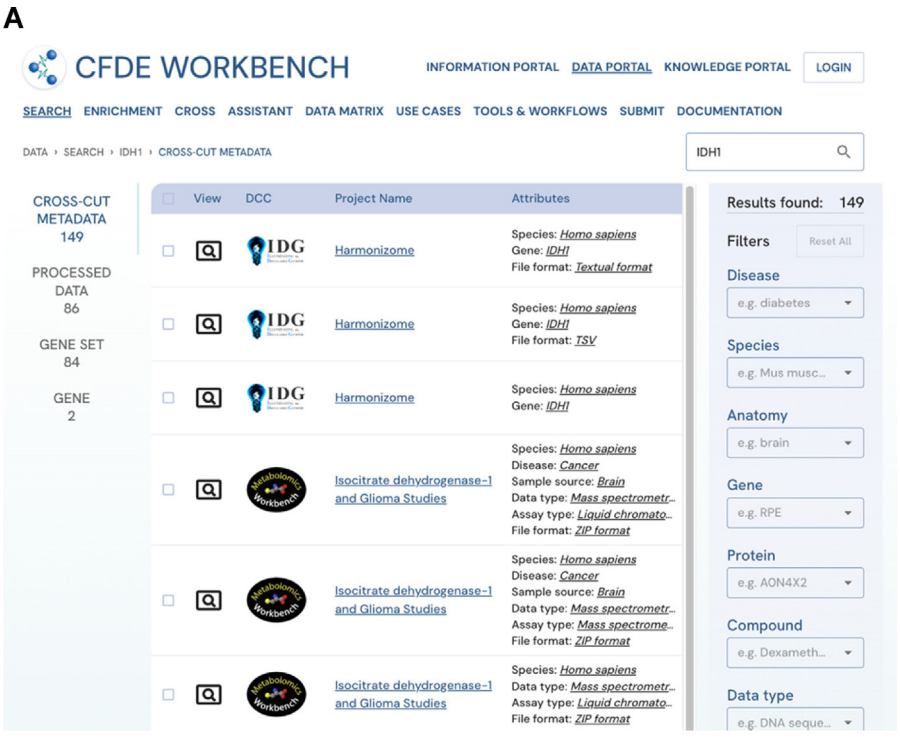


Figure 11. (A) Screenshot from the C2M2 search results when searching for IDH1 using the Data Portal search engine. (B) Screenshot from the Playbook Workflow Builder when querying the CFDE Workbench AI Assistant with the query “In which GTEx tissues is IDH1 expressed?”.

links to tools developed by the CFDE consurtium to analyze IDH1 further. These include the PWB [44], the CFDE Gene-Centric Appyter [67], and the GDLPA [48], a resource that provides links to many gene landing pages from external databases. When the PWB card is selected, a workflow is invoked to utilize knowledge from several DCCs to generate a report about IDH1.

The report contains expression levels across tissues from GTEx [1], drugs and single gene knockouts from cell lines that maximally increase or decrease the expression of IDH1 based on LINCS L1000 data [57], mouse phenotypes from KOMP2 [68], associated metabolites, pathways, and reactions from Metabolomics Workbench (MW) [5], glycosylation sites from GlyGen [6], and associated cell type from HuBMAP [8]. Given that IDH1 is an isozyme that catalyzes the reversible oxidative decarboxylation of isocitrate to α-ketoglutarate (α-KG) while reducing NADP+ to NADPH, the PWB workflow programmatically accesses MetGENE [69]. The MetGENE API returns metabolomics studies from MW [5] concerning IDH1, focusing on the pathways and reactions in which IDH1 participates, as well as the metabolites of those reactions. Genetic variants in IDH1 mutation, such as R132H, R132C, R132G, R132S, and R132L that are known to be sensitive to Ivosidenib, which inhibits D-2-hydroxyglutarate production, are obtained from the ClinVar [52] gene set library from Enrichr [70].

The CFDE Workbench AI Assistant can be used to provide similar knowledge about IDH1. For example, if a user wants to investigate which tissue types highly express IDH1, they can submit the question “In which GTEx tissues is IDH1 expressed?”. The assistant will then return a bar plot that displays the expression of IDH1 across 54 human GTEx tissues. The user can also view this information in the PWB [44] by clicking the “Open in the Playbook Workflow Builder“ button (Figure 11B). This information could help in choosing a model system to study IDH1 via genetic or chemical perturbations. Next, the user may be interested in examining differential gene expression induced by IDH1 knockouts, knockdowns, and overexpression, as well as potential drugs and pre-clinical compounds that may increase or decrease IDH1 expression. This information can be obtained from the LINCS resources [71] by submitting to the CFDE AI Assistant the question “Which LINCS drugs would maximally increase the expression of IDH1?”. The assistant returns a volcano plot and a table with the top LINCS compounds. Such information can be used for drug and target discovery. Thus, the CFDE Workbench suite of tools can enable biomedical researchers to access various resources under one dashboard, enabling them to examine the role of IDH1 in multiple contexts. By bringing together digested knowledge about genes and proteins from diverse CF resources, a more unified view of the collective role of each gene, and their protein product, could emerge.

The CFDE Workbench cloud infrastructure

The CFDE Workbench infrastructure is hosted within an Amazon Web Services (AWS) cloud account. Different components of the workbench, for example, the Information Portal, the Data Portal and the chatbot, are served by different Docker containers orchestrated by Kubernetes [72].

Conclusions and Future Directions

By combining the data and information portals (Tables 4- information, and 5-data), the CFDE Workbench serves as a comprehensive resource, where users can collect both information and data from CFDE and CF resources, as well as query disease, gene, drug, and other biological entities across harmonized (meta)data from each CF DCC. For example, for C2M2, the search and filtering may support cohort selection for biomedical researchers. Most importantly, as the CFDE Workbench is expected to continue to evolve over the next few years, the CFDE Workbench interface will be extended to support new datasets and tools, as well as integrate information and knowledge with the other four CFDE centers, making it a valuable resource for promoting FAIR data analyses across CF programs. There are several specific areas where the CFDE Workbench can be improved. Within the Data Portal, AI readiness can be improved by converting data into data frames that can be easily loaded into AI workflows and by making the data accessible via API to enable chatbots and other applications to utilize these APIs. The C2M2 has limited ability to capture patient metadata from CF programs such as GTEx, Kids First, and Bridge2AI. For researchers who work with protected health information (PHI), developing interoperability with databases such as the database for Genotypes and Phenotypes (dbGaP) [73] and platforms such as CAVATICA [74] can enhance the utility of the workbench. We intend to expand the C2M2 to include patient metadata such as body mass index (BMI), smoking status, and other non-identifying health information. The Data Portal’s “Tools and Workflows” page could be improved by adding more linked analysis tools and providing a directory to guide users in selecting the most applicable tools for their data. Lastly, converting C2M2 data from its current relational database to a KG database, such as Neo4j, may provide a better fit for the current metadata model; this effort is already underway. Within the Information Portal, possible areas of improvement include increasing collaboration with similar programs, producing micro-publications that detail each use case,

Table 4 Pages that are hosted by the CFDE Workbench Information Portal.

Portal	Feature	Description	URL
Information Portal	Landing Page	The information portal contains links to individual DCCs and CF partnership programs, as well as information about webinars, training, outreach programs, and CFDE documentation.	https://info.cfde.cloud/
	CF Programs	The CF Program page provides information about the Data Coordination Centers (DCCs) that are participating in the CFDE.	https://info.cfde.cloud/dcc
	Partnerships	The Partnerships page provides information about past and present CFDE partnerships. This includes descriptions, participating DCCs, and related links to publications and other products of the partnerships.	https://info.cfde.cloud/partnerships
	Training & Outreach	The training and outreach activities hosted by CFDE centers and DCCs are provided. Filters are available to help find relevant events, and CF community members can request to add events using a form.	https://info.cfde.cloud/training_and_outreach
	Publications	The publications under CF grants are provided along with associated DCC tags, links, and downloadable citations.	https://info.cfde.cloud/publications
	Webinars	The Webinars page features CFDE's upcoming and past webinars with their title, presenters, summaries, and video links.	https://info.cfde.cloud/training_and_outreach/cfde-webinar-series
	Documentation	The Documentation page contains descriptions of CFDE consortium standards, protocols, user guides, and other documents generated by the CFDE centers, partnerships, and DCCs.	https://info.cfde.cloud/documentation

Table 5 Pages that are hosted by CFDE Workbench Data Portal.

Portal	Feature	Description	URL
Data Portal	Landing Page	The data portal is home to the data matrix, search engine, chatbot, tools, and use cases.	https://data.cfde.cloud/
	Search engine	The search engine provides a user-friendly method of searching CFDE metadata and processed data.	https://data.cfde.cloud/
	Enrichment	The enrichment analysis tab of the Data Portal provides direct access to CFDE- (Gene Set Enrichment) GSE.	https://data.cfde.cloud/enrichment
	Cross	The cross feature of the Data Portal provides the CFDE DCC gene set libraries crossing feature that is available in GeneSetCart [47].	https://data.cfde.cloud/cross
	Chatbot	The chatbot answers queries about DCC data types as well as general questions about the CFDE and the DCCs. It utilizes OpenAI Responses API and workflows registered in the Playbook Workflow Builder [44].	https://data.cfde.cloud/chat
	Data matrix	The Data Matrix consolidates all metadata and processed data produced by the CFDE-partnered DCCs.	https://data.cfde.cloud/matrix
	Use Cases	Lists use cases created with CFDE tools to demonstrate the capability of crossing multiple DCC assets to answer biomedical research questions and form hypotheses.	https://data.cfde.cloud/usecases
	Tools & Workflows	Lists CFDE-developed tools and workflows.	https://data.cfde.cloud/tools_and_workflows
	Data and Metadata Submission System	Provides mechanisms to upload processed data, metadata, and URLs to code assets by DCC representatives.	https://data.cfde.cloud/submit
	Fair Assessment	Automated FAIR assessments with FAIRshake [24] of C2M2, XMT, ETL, KG assertions, and code assets upon asset submission. The FAIRshake insignia is embedded near each asset to assist DCCs with improving the FAIRness of their data, metadata, and other digital assets.	https://info.cfde.cloud/documentation/FAIRshake

and expanding the submission system to enable DCCs and the CFDE centers to submit events, publications, and use cases directly to the portal. So far, the CFDE consortium has developed 25 use cases and 9 tools and workflows listed on the CFDE Workbench. It is expected that this collection will continually expand in the coming years. Overall, the CFDE Workbench is a FAIR compliant resource that enables CF data and metadata discovery, access, and analysis. The CFDE Workbench is also starting to provide interoperability with other publicly available data resources and tools such as the NCBI Gene Expression Omnibus (GEO) [75], Clinical Proteomic Tumor Analysis Consortium (CPTAC) [76], Harmonizome [77], and UniProt [78]. Automated signatures computed from GEO to create the RummaGEO [79] resource can be now analyzed with the tool developed by the DRC for the CFDE Workbench enabling hypothesis generation. Some of the CFDE processed dataset are also served and are available from Harmonizome [77], enabling interoperability with many other non-CF-supported resources. Gene and protein pages on the CFDE Workbench point to UniProt [78] protein pages, and members of the UniProt team are using the Knowledge Graph UI interface [80] that was developed by the DRC for a CFDE partnership project.

CRediT authorship contribution statement

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Heesu Kim: Visualization, Software, Project administration, Data curation. **Giacomo B. Marino:** Writing – review & editing, Visualization, Software, Methodology, Formal analysis, Data curation. **Nasheath Ahmed:** Visualization, Software, Data curation. **Srinivasan Ramachandran:** Writing – review & editing, Software, Investigation, Formal analysis. **Shankar Subramaniam:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Avi Ma'ayan:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

DATA AVAILABILITY

Code and data are publicly and freely available.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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